	Direction: force on wire X is towards wire Y (right).
5(a)	E.m.f. of a source is the amount of energy converted from non-electrical to electrical per unit charge flowing through the source.
5(b)(i)	Effective resistance of the two parallel resistors S and T, $R_{eff} = \left(\frac{1}{200} + \frac{1}{300} \right)^{-1} = 120\ \Omega$ Using $V = E - Ir$, when switch is open, $I = 0$ $V = E =$ **12.0 V** When switch is closed, $V = I R_{eff} \rightarrow I = \frac{V}{R_{eff}} = \frac{10.8}{120} = 0.0900\ \text{A}$ Using $V = E - Ir$, $r = \frac{E - V}{I}$ $= \frac{12.0 - 10.8}{0.0900}$ $= 13.33$ $\approx 13.3\ \Omega$
5(b)(ii)	Energy dissipated in the power supply

No.	Solution
	$E_{\mathrm{r}} = P_{\mathrm{r}}\ t$ $= I^2 r\ t$ $= (0.0900)^2 (13.33)(5.0 \times 60)$ $= 32.39$ $\approx 32\ \mathrm{J}$
5(b)(iii)	As temperature of the thermistor increases, its resistance decreases. The effective resistance of the thermistor and resistor S in parallel decreases. EITHER By potential divider principle, the potential difference across the parallel network of thermistor and resistor S decreases. Hence, the voltmeter reading decreases. OR The total circuit resistance decreases. Since the e.m.f. E remains the same, the current I in the circuit increases. Using $V = E - Ir$, the terminal p.d. V decreases.
6(a)	There are currently no massive bodies, such as black holes and neutron stars in the Solar System, capable of emitting gravitational waves which are detectable